

The Physical CNOT Gate, via iSWAP: Realization of Quantum Computing Theory with Superconducting Circuits

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1 Introduction

1.1 Why quantum computing?

In recent decades, quantum computing has risen to prominence as one of the primary groundbreaking technological advancements of the burgeoning Second Quantum Revolution [1], in which much more of the power and "strangeness" of quantum mechanical systems is beginning to be harnessed. Though we are still in the era of noisy intermediate-scale quantum (NISQ), quantum processors hold the potential to upend the way we look at certain problems in computation [2, 3]. Due to the fundamentally different nature - simultaneously analog and digital, and incorporating such fascinating behaviors as entanglement and interference - of quantum computing, there are areas in which it can hold massive advantages over classical computing [4, 2]. This has been demonstrated in examples like the Grover algorithm [5], which provides quadratic speedup in unstructured search and thus potential advantages in the data processing scene, or Shor's algorithm [6], which provides exponential speedup in integer factorization and thus potential to break the ubiquitous RSA encryption considered effectively unbreakable by classical computers (i.e. infeasible timescale). Quantum computing could even, as renowned physicist and Nobel laureate Richard Feynman [7] postulated, simulate quantum mechanical systems at degrees of complexity far greater than classical computers could hope to achieve [8], which would revolutionize fields like biophysics and materials science. It is this potential that has driven the field of quantum computing's rapid development [3], drawing great interest and investment from academic, government, and industry sectors alike.

1.2 Why superconducting circuits?

One of the leading physical platforms for implementing quantum computing is the superconducting circuit [9, 10, 11]. Here, a qubit is constructed as an anharmonic oscillator from circuit elements, and operations are carried out through microwave pulses and capacitive coupling. Superconducting circuits offer various advantages over other approaches like ultracold atoms [12], trapped ions [13, 14], or quantum dots [15, 16]. First, they are macroscopic and engineerable with standard lithography (semiconductor manufacturing technique(s)) [9], greatly simplifying the design, fabrication, and scaling process as well as improving compatibility with existing (classical) systems. Also, superconducting qubits are highly tunable, with energy-level spectra able to be adjusted via circuit element parameters (i.e. to serve as "artificial atoms") [10, 9], allowing a variety of different qubit properties and operations. Moreover, the microwave control and strong qubit coupling that drives gates in

superconducting circuits mean that gates are very fast - in the nanosecond range of speed. Thus, while superconducting circuits still face challenges with noise and decoherence, and require complex cryogenic systems to reach the ultra-low temperatures needed for zero electrical resistance (i.e. to superconduct), they remain one of the most promising avenues for physical implementation of quantum computing, being the first-choice system of big-name industry players like Google, IBM, and Rigetti [9].

1.3 Why CNOT?

The CNOT gate flips a target qubit depending on the state of a control qubit. As "the" primary two-qubit entangling gate, it is fundamental in quantum computing theory [2]. CNOT is crucial for many, if not all, of the most advantageous quantum algorithms [2], and plays a significant role in quantum error correction [17]. Furthermore, as Barenco et al. [18] demonstrated, CNOT has an essential role in that it forms a universal gate set when used alongside single-qubit gates - that is, *any* quantum operation can be constructed from just CNOT and single-qubit gates. As such, the physical specifics of how to implement CNOT in a particular quantum computing platform are crucial to its viability and/or efficacy.

1.4 Roadmap

In this paper, we will first build a superconducting qubit and describe how single-qubit gates are driven by microwave pulses. Then, we will cover the implementation of two-qubit gates, starting with iSWAP and then constructing CNOT from iSWAP. At each step, we will include physical explanations, mathematical descriptions, and verification of results via QuTiP simulations.

2 A Qubit from a Circuit

2.1 The LC oscillator: starting point

We begin from the classical LC oscillator, one of the simplest oscillating electrical circuits, consisting of a capacitor C and inductor L . As done in Krantz et al. [9], the classical Hamiltonian H - the function representing the energy of a system, used to describe time evolution - of an LC circuit is derived as such:

Defining flux as

$$\Phi(t) = \int_{-\infty}^t V(t') dt', \quad (1)$$

and knowing that the capacitor's/inductor's energy is

$$E(t) = \int_{-\infty}^t V(t') I(t') dt', \quad (2)$$

for voltage $V(t')$ and current $I(t')$ across each element, knowing $V = L * dI/dt$ and $I = C * dV/dt$, we can then describe the energy of each element:

$$\mathcal{T}_C = \frac{1}{2} C \dot{\Phi}^2, \quad (3)$$

$$\mathcal{U}_L = \frac{1}{2L} \Phi^2. \quad (4)$$

The Lagrangian, or difference between kinetic and potential energies, is then defined here by arbitrarily labeling the capacitor's electrical energy as kinetic and the inductor's magnetic energy as potential:

$$\mathcal{L} = \mathcal{T}_C - \mathcal{U}_L = \frac{1}{2} C \dot{\Phi}^2 - \frac{1}{2L} \Phi^2. \quad (5)$$

The way to convert the Lagrangian to the Hamiltonian is called the Legendre transformation, and in its general form looks like:

$$H = \sum p_i \dot{q}_i - \mathcal{L}, \quad (6)$$

where $\dot{q}$ is a generalized velocity and p a generalized momentum. For our purposes, charge takes the role of momentum and the time derivative of flux takes the role of velocity, which makes sense because momentum should be equal to the partial derivative of the Lagrangian with respect to velocity, and

$$Q = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C \dot{\Phi}. \quad (7)$$

Thus, we have the system's Hamiltonian as

$$H = Q \dot{\Phi} - \mathcal{L} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} \equiv \frac{1}{2} C V^2 + \frac{1}{2} L I^2, \quad (8)$$

which aligns with the standard definitions of an LC circuit.

Moving from here to a quantum Hamiltonian/description of the LC circuit, we shift Φ and Q to being commuting operators, and we introduce the dimensionless analogs of flux and charge: reduced flux $\phi \equiv 2\pi \frac{\Phi}{\Phi_0}$, with normalizing factor $\Phi_0 = \frac{h}{2e}$ (the smallest possible unit of flux) and reduced charge $n = \frac{Q}{2e}$, the number of Cooper pairs, bound pairs of electrons that move together with no resistance through superconducting circuits (and are in fact the basis of why they work), on a "circuit island," or distinct section of the circuit. Also note that a quantum Hamiltonian is no longer a function like a classical Hamiltonian, but an operator! We therefore write

$$H = 4E_C n^2 + \frac{1}{2} E_L \varphi^2, \quad (9)$$

with charging energy $E_C = \frac{e^2}{2C}$ being the energy needed to add one electron of a Cooper pair to the island, and inductive energy $E_L = \frac{(\Phi_0/2\pi)^2}{L}$.

Solving this equation for time evolution gives that the energy oscillates between the capacitor's electric field and the inductor's magnetic field at angular frequency $\omega_r = \frac{1}{\sqrt{LC}}$, with an infinite series of eigenstates with equidistantly spaced eigenenergies. The Hamiltonian can thus be rewritten as that of a quantum harmonic oscillator with ladder operators:

$$H_{LC} = \hbar \omega_r \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right), \quad (10)$$

where ladder operators $a^\dagger$ and a , respectively, represent the creation and annihilation operators, which act on the energy eigenstates $|n\rangle$ (called Fock states) to go "up" or "down" the "ladder" of energies, as such:

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle, \quad (11)$$

$$a |n\rangle = \sqrt{n} |n-1\rangle. \quad (12)$$

This creates an energy spectrum $E_n = \omega_r(n + 1/2)$, a perfectly equally spaced "ladder" where "rungs" of energy are spaced ω_r apart.

However, to use this system as a qubit, we cannot have the energy levels evenly spaced! The way superconducting qubits are transitioned between $|0\rangle$ and $|1\rangle$ are via microwave pulsing, and for evenly spaced energy levels a microwave tuned to the $|0\rangle \rightarrow |1\rangle$ transition would be equally resonant with subsequent transitions ($|1\rangle \rightarrow |2\rangle$, $|2\rangle \rightarrow |3\rangle$, etc.). Thus, binary nature would be lost as energy would just go straight up the ladder.

2.2 The Josephson junction

To get an anharmonic ladder, which does not have uniform spacing of energy levels, we replace the inductor with a Josephson junction - two superconductors separated by a thin barrier. The junction's energy is a nonlinear function: $-E_J \cos(\phi)$. Transmons (transmission-line shunted plasma oscillation qubits) [19] are designed such that the junction energy is much greater than the charging energy, so, after Taylor expansion of the cosine and taking the quadratic piece as well as the first nonlinear correction (the quadratic term), we get an LC -like oscillator represented by the nonlinear/anharmonic Duffing oscillator Hamiltonian

$$H_{\text{transmon}} = \hbar\omega_q a^\dagger a + \frac{\alpha}{2} a^\dagger a^\dagger a a, \quad (13)$$

with anharmonicity α being the difference between transition frequencies for $|0\rangle \rightarrow |1\rangle$ and $|1\rangle \rightarrow |2\rangle$ (see Fig. 1). Thus, a microwave drive narrower than $|\alpha|$ can flip $|0\rangle \leftrightarrow |1\rangle$ without going into higher levels.

2.3 The two-level system

Because now the higher levels are detuned by $|\alpha|$, we can simply work with pulses slower than $1/|\alpha|$ to limit the transmon to a two-level system, since those would keep the transmon within the lowest two energy levels. Thus, the qubit Hamiltonian becomes

$$H_q = \frac{\hbar\omega_q}{2} \hat{\sigma}_z, \quad (14)$$

where $\hat{\sigma}_z$ is the same Pauli operator one would be familiar with from quantum information theory [20]: $\hat{\sigma}_z |0\rangle = +|0\rangle$, $\hat{\sigma}_z |1\rangle = -|1\rangle$. Everything that follows is built on this foundation. (Of course, this is just an approximation, though - Sec. 6 goes over what happens when the approximation breaks down.) Also note that from here on, we will begin to omit the $\hbar$ and work in the realm $\hbar = 1$, since we are most concerned with the theory and the dynamics of how things work (this is just a unit change and will not affect the picture).

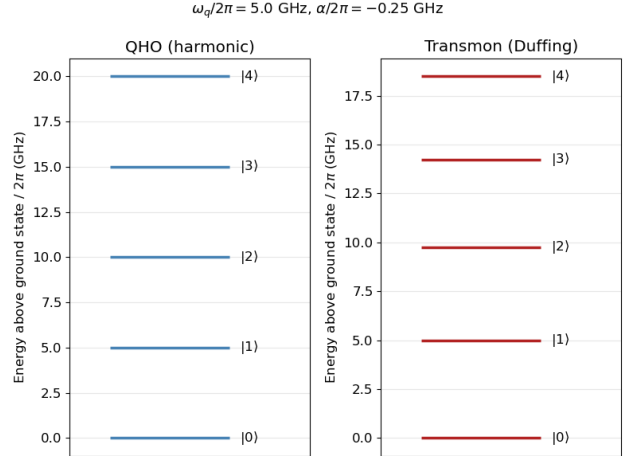


Figure 1: Energy-level ladder of the simple LC oscillator (left) compared to the transmon (right). Note the difference equal to α between the $|1\rangle \rightarrow |2\rangle$ and $|0\rangle \rightarrow |1\rangle$ transitions, which makes it possible to stay just within the bottom two levels. Generated with QuTiP.

Note that this shows that a qubit at rest precesses under $\hat{\sigma}_z$, i.e. the state's Bloch vector representation rotates about the z-axis with angular frequency ω_q . This is the base case for time evolution under a Hamiltonian: applying gates/unitaries to the circuit simply means evolving under some different Hamiltonian for some amount of time.

3 Single-Qubit Gates

3.1 The qubit drive

To drive single-qubit gates, we capacitively couple a superconducting qubit (transmon) to a microwave source with a "drive line" as in Fig. 2, applying microwave pulses resonant to the qubit's. This manifests as a transverse drive, appearing in the Hamiltonian here as the second term (in addition to the natural precession) [9]:

$$H = \frac{\omega_q}{2} \hat{\sigma}_z + \frac{\Omega}{2} \hat{\sigma}_x. \quad (15)$$

As can be seen in Fig. 3, the full picture looks like fast oscillations happening at ω_q while the big-picture time evolution happening under the drive happens at Ω . This oscillation between $|0\rangle$ and $|1\rangle$ resulting from the drive is called Rabi oscillation, and the frequency Ω - proportional to the (oscillatory) driving pulse's amplitude - is called the Rabi frequency.

Note that the oscillating drive's own frequency must first be tuned to match the qubit's precession (resonant), $\omega_q = \omega_d$! We will go over this in the two-qubit case.

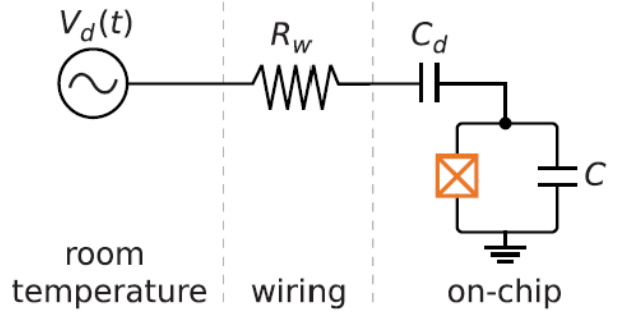


Figure 2: Circuit diagram of a microwave drive line ($V_d(t)$) capacitively coupled to a transmon (orange box with X = Josephson junction). From Krantz et al. [9]

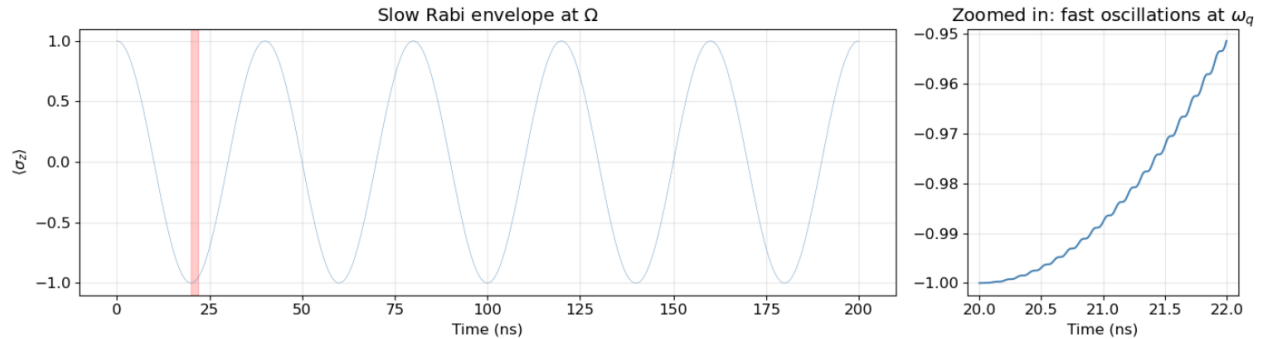


Figure 3: Large/slow Rabi oscillation(s) at Ω of a transmon under resonant drive (left); small/fast oscillations at ω_q of natural precession (right). Parameters: $\omega_q/2\pi = 5$ GHz, $\Omega/2\pi = 50$ MHz. Generated with QuTiP.

3.2 The rotating wave approximation (RWA)

One tool to focus on the dynamics resulting from the drive (and not those from the usual rest state precession) is to use a frame that rotates with the qubit at frequency ω_q to remove that fast

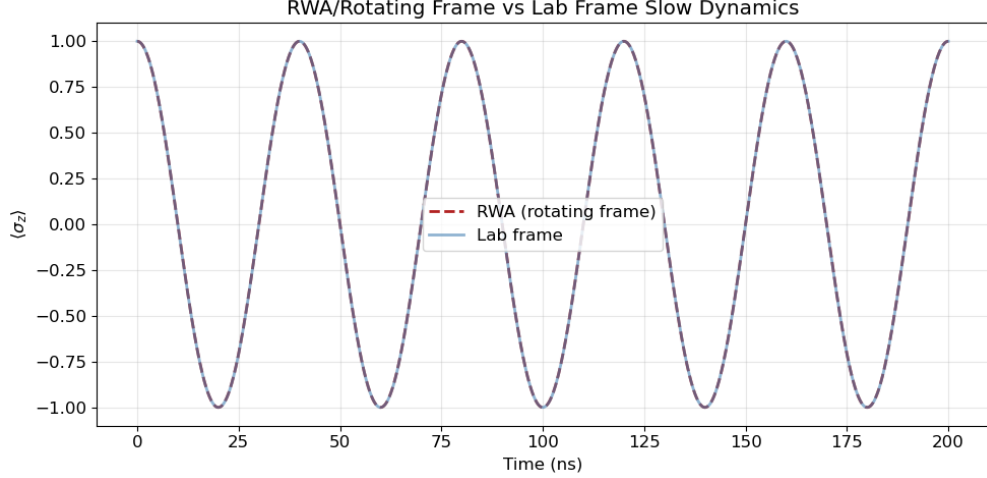


Figure 4: The rotating wave approximation (red, dashed) overlayed on top of the same slow Rabi envelope from Fig. 3; note the near-perfect match at this timescale. Generated with QuTiP.

oscillation element. The drive then can be separated into slow terms and fast terms, the latter of which can be dropped because they oscillate at $2\omega_q$ and so average to zero for timescales longer than $1/\omega_q$ - which is true for timescales of the almost all pulses used in context. This is called the rotating wave approximation (RWA), which matches very well with the simple slow Rabi envelope (see Fig. 4) and thus shows that we need not worry about the complex time-related dynamics and can just write the simple static Hamiltonian for the Rabi oscillation,

$$H_{RWA} = \frac{\Omega}{2}(\cos \phi \hat{\sigma}_x + \sin \phi \hat{\sigma}_y), \quad (16)$$

where ϕ is the microwave drive's phase.

Effectively, evolving under this Hamiltonian means rotating the Bloch vector at Ω about some direction (based on ϕ) of a static field in the XY-plane of the Bloch sphere created by this drive.

3.3 The X gate (and other single-qubit gates)

If we choose $\phi = 0$ and pulse for time T such that $\int_0^T \Omega(t) dt = \pi$, the resulting operation is a rotation of π about the x axis:

$$U = e^{\frac{iHt}{\hbar}} = e^{-i\frac{\pi}{2}\hat{\sigma}_x} = -iX. \quad (17)$$

Discarding the physically irrelevant global phase $-i$, this is exactly the single-qubit X gate!

Similarly, we could pulse for the same time T with $\phi = \pi/2$ to get Y , get Z as a "virtual Z gate" by adjusting phase definition for future X/Y pulses (with no physical pulse needed, only software!) [21], and pulse for half the time ($\int_0^T \Omega(t) dt = \frac{\pi}{2}$) along with some phase adjustments (i.e. virtual Z) to get a Hadamard. We can also achieve arbitrary rotations by simply adjusting these parameters of pulse length, phase, etc.

4 Two-Qubit Gates: iSWAP

4.1 Capacitive coupling of two transmons

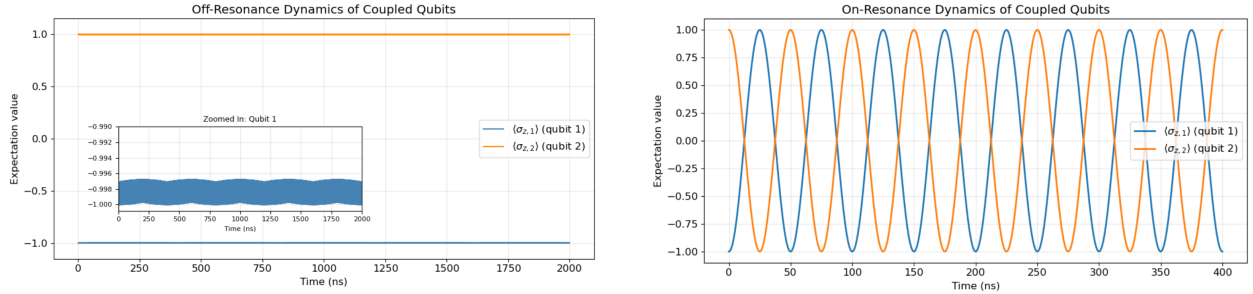
Two transmons can be linked by a coupling capacitor, which allows them to exchange AC energy (photons): the oscillating electric field(s) of one qubit can influence the other, and vice versa, allowing excitations to essentially be transferred from one qubit to the other (and back). This interaction, after applying an RWA (which, here, would remove the fast-oscillating counter-rotating terms like $\sigma_{+,1}\sigma_{+,2}$) and assuming we restrict each qubit to a two-level system, is represented by the two-qubit Hamiltonian [9]

$$H = \frac{\omega_1}{2}\sigma_{z,1} + \frac{\omega_2}{2}\sigma_{z,2} + g(\sigma_{+,1}\sigma_{-,2} + \sigma_{-,1}\sigma_{+,2}), \quad (18)$$

where $\sigma_+ = |0\rangle\langle 1|$ and $\sigma_- = |1\rangle\langle 0|$ represent the raising and lowering operators, respectively (the two-level equivalents of $a^\dagger$ and a), and g is the coupling strength (dependent on physical factors like distance, orientation, intensity of interaction, etc.). The first two terms of this Hamiltonian are the energies of each individual qubit, while the third is the one representing the coupling: $\sigma_{+,1}\sigma_{-,2}$ is an excitation moving from qubit 2 to qubit 1, and $\sigma_{-,1}\sigma_{+,2}$ is an excitation moving from qubit 1 to qubit 2.

This interaction, then, causes the transition $|01\rangle \leftrightarrow |10\rangle$ while leaving $|00\rangle$ and $|11\rangle$ alone.

Here we will also address the question first brought up in Sec. 3.1 of the need for resonance, or for the qubits' frequencies to match. In Fig. 5, we show two similarly coupled two-qubit systems - one where the two qubits - one in state $|0\rangle$, the other in $|1\rangle$ - have the same frequency ("on-resonance"), one where they don't ("off-resonance"). The off-resonance case shows a greatly suppressed interaction, whereas the on-resonance case shows the expected full excitation swap.



(a) Off-resonance case: $\Delta/2\pi = 0.5$ GHz. Greatly suppressed interaction, with inset zoomed section for qubit 1 displaying amplitude $\sim (g/\Delta)^2 = 0.00040$.

(b) On-resonance case: $\Delta = 0$. Full swap visible, with period of 50 ns (swap in half period, so 25 ns; another half period brings back to original state).

Figure 5: Off- vs On-Resonance Dynamics of Two Coupled Qubits, with $g/2\pi = 10$ MHz. Clearly, only the on-resonance case displays the desired interaction. Also note that traces for each qubit are perfectly anticorrelated with each other. Generated with QuTiP.

4.2 Resonant exchange of excitations: a mathematical description

When the two qubits *are* at the same frequency ($\omega_1 = \omega_2 = \omega_q$), the resonant coupling causes an excitation to swap back and forth in sync - i.e. coherently, going through superposition(s) of states. Eq. (18) becomes

$$H = \frac{\omega_q}{2}(\sigma_{z,1} + \sigma_{z,2}) + g(\sigma_{+,1}\sigma_{-,2} + \sigma_{-,1}\sigma_{+,2}), \quad (19)$$

and the first term, representing free precession, can be dropped by moving into the frame rotating at ω_q . The resulting Hamiltonian can be written as a 4x4 matrix, in the computational basis:

$$\tilde{H} = g(\sigma_{+,1}\sigma_{-,2} + \sigma_{-,1}\sigma_{+,2}) = g \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (20)$$

The unitary is $U(t) = e^{-iHt}$, and since H is nontrivial only in the center 2x2 block, which is equivalent to $g\hat{\sigma}_x$, we can just exponentiate that block in the $\{|01\rangle, |10\rangle\}$ subspace:

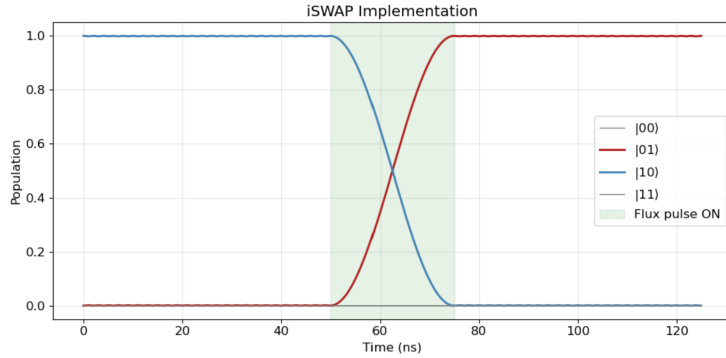
$$U(t) = e^{-igt\hat{\sigma}_x} = \cos(gt)I - i\sin(gt)\hat{\sigma}_x = \begin{pmatrix} \cos(gt) & -i\sin(gt) \\ -i\sin(gt) & \cos(gt) \end{pmatrix} \quad (21)$$

which, embedding back into the 4x4 space, becomes

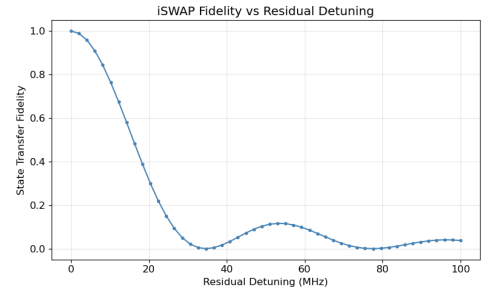
$$U(t) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos gt & -i\sin gt & 0 \\ 0 & -i\sin gt & \cos gt & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (22)$$

From here the dynamics are even clearer to see: qubit 2 rotates conditional on qubit 1's excitation, and vice versa.

4.3 The iSWAP gate



(a) Implementation of the iSWAP gate, effecting $|10\rangle \rightarrow -i|01\rangle$. Starting fully in $|10\rangle$, ending fully in $(-i)|01\rangle$. Note barely-visible small oscillations when detuned. Green region is on-resonance time period (duration 25 ns).



(b) iSWAP fidelity, measured by $|\langle 01|\psi(t^*)|01\rangle\psi(t^*)|^2$ (success of state transfer as overlap of end population and expected end state), as residual detuning increases.

Figure 6: iSWAP implementation and fidelity relative to detuning. Qubit 1 fixed at $\frac{\omega_1}{2\pi} = 5.0$ GHz, qubit 2 variable/tunable. Coupling $\frac{g}{2\pi} = 10$ MHz. Generated with QuTiP.

From the same setup as described in the previous two subsections, we can simply set pulse duration $t^* = \pi/(2g)$, which gives $\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$. This produces the iSWAP gate, which has the matrix, derived from Eq. (22),

$$U_{iSWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (23)$$

Physically, we achieve this by starting with the two qubits detuned (off-resonance case, negligible coupling interaction). Applying a flux pulse to one, we temporarily bring it into resonance with the other, wait a certain time (t^*), and then turn off the pulse to detune the qubits again. Thus, the interaction is held for a set amount of time t^* and results in the iSWAP gate (see Fig. 6a). We can discount the comparatively small amount of time needed to bring the qubits in/out of resonance, since precise frequency matches are needed. Indeed, a key concern in the physical setup is reducing residual detuning, a relatively small amount of which can result in large drops in gate fidelity (Fig. 6b - <1% detuning causes fidelity to plummet). Also note that iSWAP is perfectly entangling - this will be important for the next section!

5 Constructing CNOT from iSWAP

5.1 Circuit Construction

As stated by Schuch and Siewert [22], two applications of the iSWAP gate, combined with single-qubit rotations, produces the CNOT gate. The circuit is shown in Fig. 7.

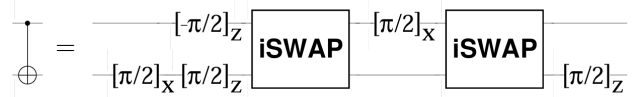


Figure 7: Decomposition of CNOT gate: implemented as two iSWAPs and various single-qubit gates. From Schuch and Siewert [22].

Now, we go over the intuition of why the decomposition works. iSWAP and CNOT differ because, although both perfectly entangling, iSWAP is symmetric and CNOT is asymmetric (i.e. between control and target qubits). Another way of thinking about this is that iSWAP always preserves the number of "1"s because the excitation oscillates between qubits, while CNOT doesn't care about that - it just flips based off the state of one qubit, the two qubits don't intercorrelate (there's phase kickback, but that's another story). Thus to bridge the gap we include a local rotation between them, which breaks the symmetry between the two qubits by effectively "twisting" the axis of the second iSWAP. This causes some cancellation between the two iSWAPs, so the total interaction becomes asymmetric. The remaining single-qubit rotations effectively change bases such that the resulting entanglement aligns with CNOT in the computational basis.

Note that iSWAP is a more "elementary" or "native" gate to the physical architecture (that is why we focused on explaining its physical nature above!), and thus CNOT is usually constructed from iSWAP and not the other way around. Also, since the circuit for CNOT includes so many more gates, this means that CNOT is inherently a more noisy gate for superconducting circuit hardware: more gates/complexity means both more room for error and more time for qubits to decohere.

5.2 Verification in QuTiP

We demonstrate that the above circuit using two iSWAPs and multiple single-qubit rotations actually produces CNOT by building the gate in QuTiP. The resulting matrix is

$$U = e^{-\frac{i\pi}{4}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (24)$$

which is exactly CNOT up to a global phase $e^{-\frac{i\pi}{4}}$. Sure enough, applying the circuit to all four computational basis states reproduces the CNOT truth table:

Input		Output	
c	t	c	$t \oplus c$
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Table 1: CNOT truth table, reproduced in QuTiP by applying decomposition involving 2x iSWAP and single-qubit rotations. Note exact match with expected values.

6 Limitations of the Two-Level Approximation: Leakage

Thus far, we have stuck to the two-level approximation for each transmon. However, this is not the full physically correct story (indeed, the real Hilbert space should be infinite dimensional, and each transmon should technically have infinite energy levels). Thus, some of the population can and will "leak" into the upper levels, beyond the computational subspace of the bottom two for each transmon - an error!

We will work here by simulating a two-qubit system where each qubit has four energy levels, for a total 16-dimensional Hilbert space. The coupling term becomes $g(a_1^\dagger a_2 + a_1 a_2^\dagger)$, using the proper ladder operators. In Fig. 8, we demonstrate that, for small g , leakage remains very small. Moreover, there is a natural "self-correcting" nature for iSWAP - for gate time t^* , the leakage is minimized (i.e. there is a node). This is because the leakage channels out of $|10\rangle$ and $|01\rangle$ go to different higher levels - $|21\rangle$ and $|12\rangle$, respectively - and with different phases, as the swap happens. Thus, the leakage from the first half of the iSWAP interaction interferes destructively with that from the second half, so after one full iSWAP duration total leakage is at a minimum.

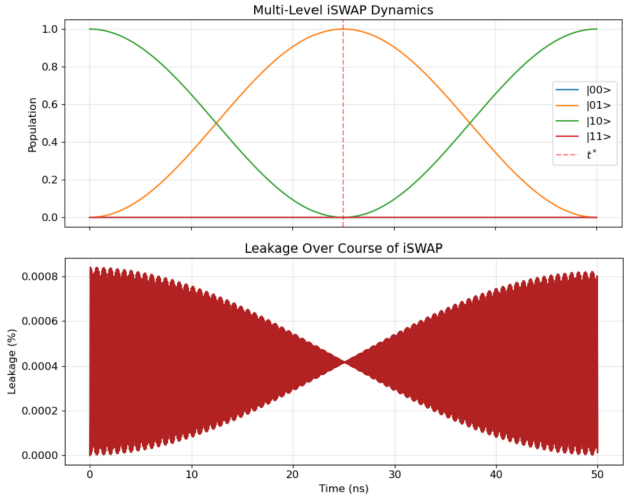


Figure 8: Leakage over time in an iSWAP gate: relatively small magnitude for this $g/2\pi = 10$ MHz: max amplitude at approx 0.0008% of total population. Fast oscillations from counterrotating terms $a_1 a_2$ and $a_1^\dagger a_2^\dagger$, negligible and can be removed via a rotating frame/RWA; slow envelope significant, from overall swap dynamics. Note minimized leakage at exactly one iSWAP duration t^* . Generated in QuTiP.

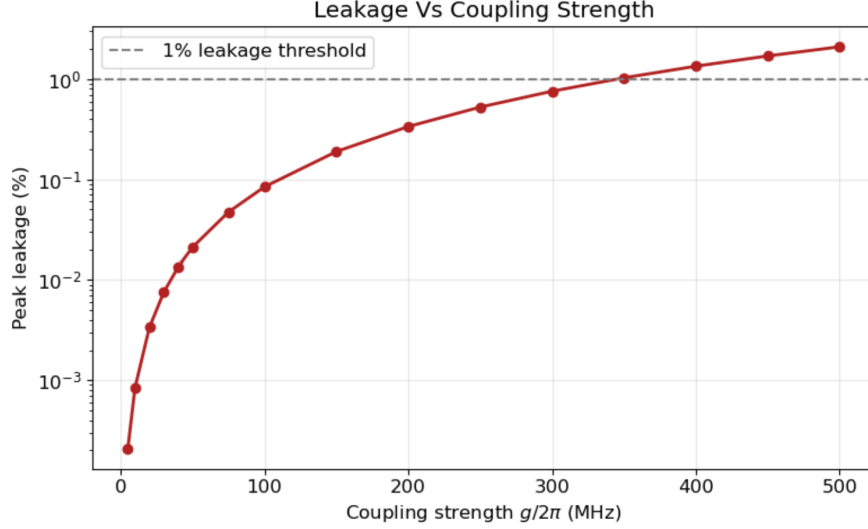


Figure 9: Max leakage vs coupling strength g . Positive correlation; greater coupling strength (faster gate speed) causes increased leakage. Crosses 1% leakage threshold at $g/2\pi \approx 350$ MHz. Note that here $\omega_q = 5.0$ GHz, so g is 7% of ω_q when it gets to 1% leakage. Generated with QuTiP.

However, also of note is that leakage scales like $(g/\omega_q)^2$, so it is crucial to keep $g \ll \omega_q$ - Fig. 9 showcases this relationship. Since leakage and coupling strength are correlated, there is an inherent tradeoff between gate speed (which is inversely proportional to coupling strength) and fidelity.

7 Outlook and Conclusion

7.1 Future Directions

Expanding on Section 5.1's brief mention of CNOT being a noisy gate, the iSWAP construction of CNOT is a costly one. Effectively, since two iSWAPs must be used to produce one CNOT, CNOT creates double the loss in circuit fidelity as a two-qubit gate (plus whatever, albeit lower, levels of noise are introduced by the single-qubit gates. Thus, a big question in the field is how to achieve a less noisy CNOT.

One approach is the CPHASE gate, of which only one instance is needed (plus single-qubit gates) to produce CNOT. CPHASE enacts a phase shift on a target qubit based on the state of a control qubit, actually by driving the system's state to interact with higher energy levels (i.e. $|11\rangle$ interacting with $|20\rangle$) [23] beyond the computational subspace (thus taking advantage of regions which would be leakage errors in the iSWAP case) [9]. Thus, advancements in refining CPHASE trajectories - the path(s) taken to achieve the phase shift - and waveform calibration could lead to much higher CNOT fidelity from a CPHASE construction.

Additionally, cross-resonance (CR) is a completely different approach [24, 25, 26], using fixed-frequency qubits where one qubit is driven at the frequency of the other. The effective Hamiltonian picks up a ZX term that entangles the qubits, which can be used to synthesize CNOT. This reduces flux noise coming from tuning qubits, since the qubits' frequencies don't change and only microwave drive(s) are used. One of the challenges with the CR approach is that it has an always-on residual qubit coupling which can create unwanted ZZ interactions between qubits while the gate is not active, leading to decoherence/fidelity drops. Thus, the focus in this field is on canceling out this coupling actively with techniques like pulse sequences (e.g. echo-CR [27]) or tunable couplers.

Indeed, modern approaches have been shifting particularly toward adoption of tunable couplers and parametric gates [9], which provide high-fidelity, fast gates with lower decoherence. Instead of a static/passive coupler, tunable couplers utilize additional circuit elements (like superconducting quantum interference devices - SQUIDs) that can be modulated via applying external magnetic flux to turn the interaction between qubits on and off, minimizing residual coupling. Parametric gates modulate the frequencies of tunable couplers via fast magnetic flux pulses to cause resonant interactions that generate entangling gates - since the interaction only happens when modulation is occurring, this offers a higher degree of control and avoids flux noise since only the coupling element is targeted.

All these directions - and more, such as those relating to materials/fabrication, logical CNOT implementations that are much more fault-tolerant than the base case - could lead to much more efficient and effective implementation of the CNOT gate. This in turn would mean faster and/or greater fidelity quantum operations in general, due to the ubiquitous nature of CNOT in quantum computing.

7.2 Summary

In this paper, we walked through the physical implementation of various quantum computing theoretical constructs in superconducting circuits, leading up to the CNOT gate. We began from a simple LC oscillator, introduced the Josephson junction to restrict the system to two levels (the transmon), and implemented single-qubit gate(s) via microwave drive. Then, we described how two-qubit gates can be achieved through capacitive coupling of transmons, starting with iSWAP and then using that to construct CNOT. Throughout the process, we described the physics of each system with Hamiltonians, QuTiP simulations, etc., and we linked these physical ideas to quantum theory constructs like unitaries. Furthermore, we touched upon implementation issues like fidelity and leakage. Thus, we have described the physical foundation for some of the key components of the superconducting circuit platform for quantum computing.

8 Acknowledgements and Code Availability

Special thanks to Teaching Fellow Gautham Umasankar for creating a pset (with the aid of AI tools) that taught me many of the concepts covered in this paper, as well as guided the decision and coding process to create the figures. Code segments from QuTiP [28] documentation and public resources were also largely utilized/adapted for the simulations and figure creation. A "cleaned" (i.e. trimmed to include only what was used in figure creation) copy of the code has been provided in an attached PDF as a Jupyter notebook, as well as a copy of the pset.

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